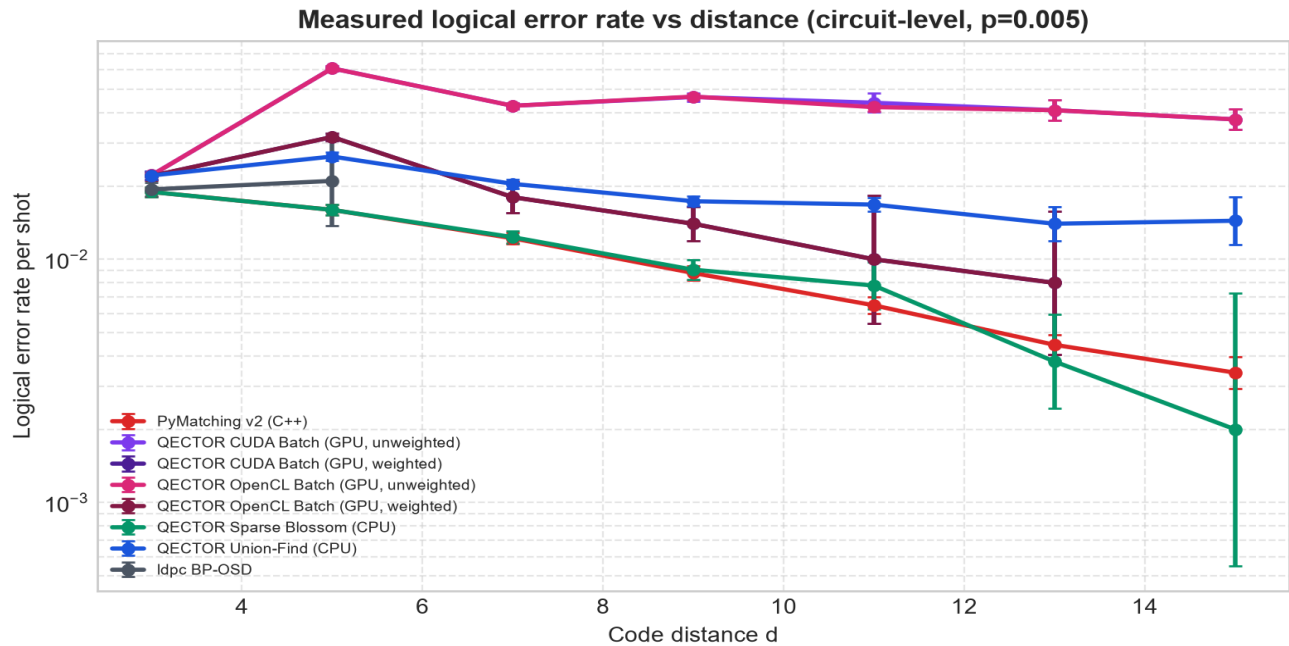
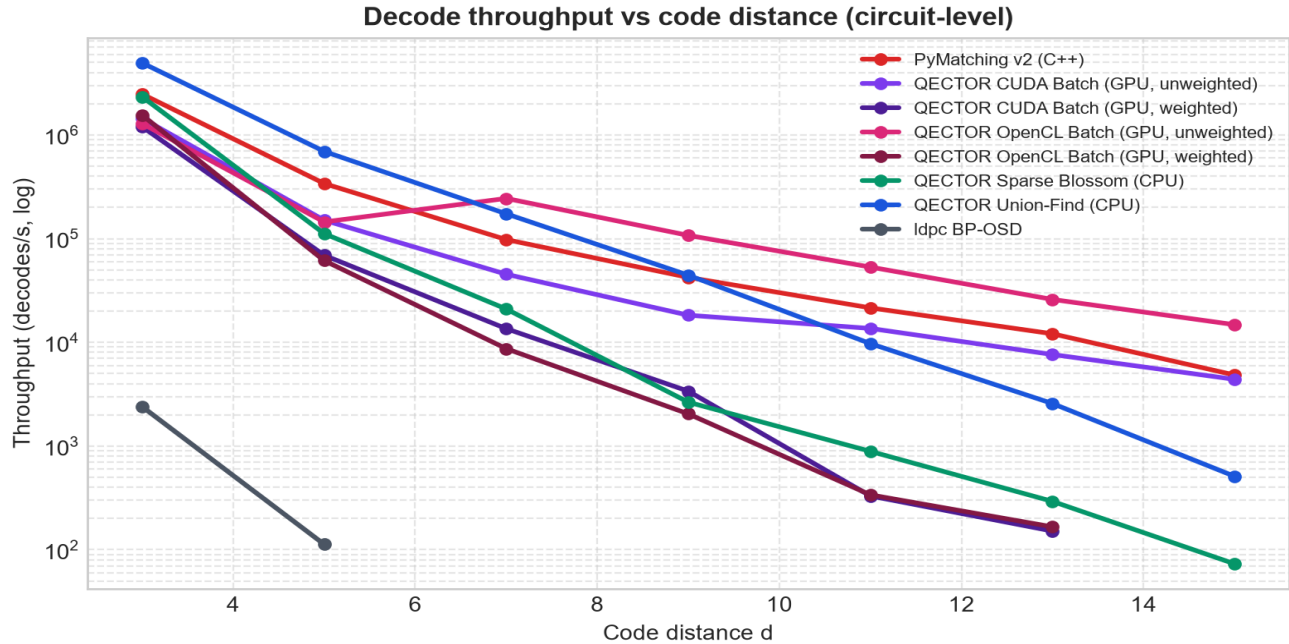


QECTOR v0.7.0 — circuit-level decoder comparison

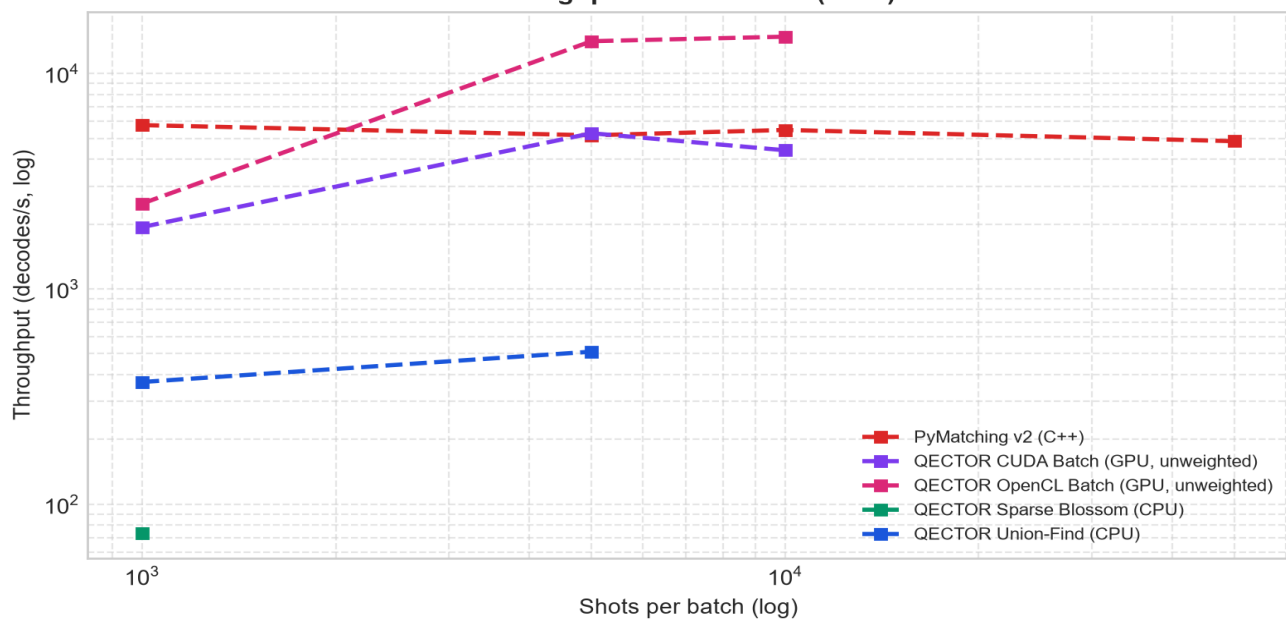
Measured with `ler.estimate_ler_circuit_level`: identical circuit, DEM, samples and decode_batch resolver for every decoder, scored against the circuit's logical observables. Validated by `ler.assert_comparable`. No extrapolated cells.

Noise model: circuit-level, $p=0.005$ | Commit: b436f04e3b | Generated: 2026-08-01T04:33:23Z

Platform: Windows-10-10.0.26100-SP0 | Python: 3.11.9



Throughput vs batch size (d=15)



Measured results (187 cells)

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
PyMatching v2 (C++)	3	1,000	13	0.01300	[0.0076, 0.0221]	2,059,308	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	1,243,936	0.60x
QECTOR CUDA Batch (GPU, weighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	1,014,816	0.49x
QECTOR OpenCL Batch (GPU, unweighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	477,623	0.23x
QECTOR OpenCL Batch (GPU, weighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	323,300	0.16x
QECTOR Sparse Blossom (CPU)	3	1,000	13	0.01300	[0.0076, 0.0221]	706,564	0.34x
QECTOR Union-Find (CPU)	3	1,000	16	0.01600	[0.0099, 0.0258]	1,077,935	0.52x
Idpc BP-OSD	3	1,000	12	0.01200	[0.0069, 0.0209]	2,411	0.001x
PyMatching v2 (C++)	3	5,000	84	0.01680	[0.0136, 0.0208]	2,387,091	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,364,443	0.57x
QECTOR CUDA Batch (GPU, weighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,288,892	0.54x
QECTOR OpenCL Batch (GPU, unweighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,409,125	0.59x
QECTOR OpenCL Batch (GPU, weighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,242,977	0.52x
QECTOR Sparse Blossom (CPU)	3	5,000	84	0.01680	[0.0136, 0.0208]	1,526,298	0.64x
QECTOR Union-Find (CPU)	3	5,000	105	0.02100	[0.0174, 0.0254]	2,156,939	0.90x
Idpc BP-OSD	3	5,000	87	0.01740	[0.0141, 0.0214]	2,320	0.001x
PyMatching v2 (C++)	3	10,000	202	0.02020	[0.0176, 0.0231]	2,414,176	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	10,000	235	0.02350	[0.0207, 0.0267]	2,027,370	0.84x
QECTOR CUDA Batch (GPU, weighted)	3	10,000	238	0.02380	[0.0210, 0.0270]	1,444,961	0.60x
QECTOR OpenCL Batch (GPU, unweighted)	3	10,000	235	0.02350	[0.0207, 0.0267]	1,681,775	0.70x
QECTOR OpenCL Batch (GPU, weighted)	3	10,000	238	0.02380	[0.0210, 0.0270]	1,311,751	0.54x
QECTOR Sparse Blossom (CPU)	3	10,000	202	0.02020	[0.0176, 0.0231]	1,918,686	0.80x
QECTOR Union-Find (CPU)	3	10,000	233	0.02330	[0.0205, 0.0264]	3,129,597	1.30x
Idpc BP-OSD	3	10,000	199	0.01990	[0.0173, 0.0228]	2,355	0.001x
PyMatching v2 (C++)	3	50,000	960	0.01920	[0.0180, 0.0204]	2,483,337	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	50,000	1131	0.02262	[0.0214, 0.0240]	982,560	0.40x
QECTOR CUDA Batch (GPU, weighted)	3	50,000	1129	0.02258	[0.0213, 0.0239]	1,158,896	0.47x
QECTOR OpenCL Batch (GPU, unweighted)	3	50,000	1131	0.02262	[0.0214, 0.0240]	1,192,999	0.48x
QECTOR OpenCL Batch (GPU, weighted)	3	50,000	1129	0.02258	[0.0213, 0.0239]	1,340,210	0.54x
QECTOR Sparse Blossom (CPU)	3	50,000	960	0.01920	[0.0180, 0.0204]	2,285,098	0.92x
QECTOR Union-Find (CPU)	3	50,000	1128	0.02256	[0.0213, 0.0239]	4,478,521	1.80x
Idpc BP-OSD	3	50,000	969	0.01938	[0.0182, 0.0206]	2,399	0.001x
PyMatching v2 (C++)	3	100,000	1891	0.01891	[0.0181, 0.0198]	2,478,782	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	100,000	2215	0.02215	[0.0213, 0.0231]	1,446,123	0.58x
QECTOR CUDA Batch (GPU, weighted)	3	100,000	2201	0.02201	[0.0211, 0.0229]	1,201,288	0.48x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR OpenCL Batch (GPU, unweighted)	3	100,000	2215	0.02215	[0.0213, 0.0231]	1,286,159	0.52x
QECTOR OpenCL Batch (GPU, weighted)	3	100,000	2201	0.02201	[0.0211, 0.0229]	1,554,014	0.63x
QECTOR Sparse Blossom (CPU)	3	100,000	1891	0.01891	[0.0181, 0.0198]	2,337,448	0.94x
QECTOR Union-Find (CPU)	3	100,000	2210	0.02210	[0.0212, 0.0230]	4,953,928	2.00x
PyMatching v2 (C++)	5	1,000	19	0.01900	[0.0122, 0.0295]	295,203	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	1,000	60	0.06000	[0.0469, 0.0765]	86,730	0.29x
QECTOR CUDA Batch (GPU, weighted)	5	1,000	31	0.03100	[0.0219, 0.0437]	36,515	0.12x
QECTOR OpenCL Batch (GPU, unweighted)	5	1,000	60	0.06000	[0.0469, 0.0765]	127,962	0.43x
QECTOR OpenCL Batch (GPU, weighted)	5	1,000	31	0.03100	[0.0219, 0.0437]	36,462	0.12x
QECTOR Sparse Blossom (CPU)	5	1,000	19	0.01900	[0.0122, 0.0295]	89,467	0.30x
QECTOR Union-Find (CPU)	5	1,000	26	0.02600	[0.0178, 0.0378]	320,626	1.09x
ldpc BP-OSD	5	1,000	21	0.02100	[0.0138, 0.0319]	113	0x
PyMatching v2 (C++)	5	5,000	73	0.01460	[0.0116, 0.0183]	322,585	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	5,000	336	0.06720	[0.0606, 0.0745]	215,810	0.67x
QECTOR CUDA Batch (GPU, weighted)	5	5,000	181	0.03620	[0.0314, 0.0417]	106,719	0.33x
QECTOR OpenCL Batch (GPU, unweighted)	5	5,000	336	0.06720	[0.0606, 0.0745]	235,303	0.73x
QECTOR OpenCL Batch (GPU, weighted)	5	5,000	181	0.03620	[0.0314, 0.0417]	82,006	0.25x
QECTOR Sparse Blossom (CPU)	5	5,000	73	0.01460	[0.0116, 0.0183]	106,067	0.33x
QECTOR Union-Find (CPU)	5	5,000	148	0.02960	[0.0253, 0.0347]	536,815	1.66x
PyMatching v2 (C++)	5	10,000	166	0.01660	[0.0143, 0.0193]	319,172	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	10,000	600	0.06000	[0.0555, 0.0648]	240,079	0.75x
QECTOR CUDA Batch (GPU, weighted)	5	10,000	316	0.03160	[0.0283, 0.0352]	90,403	0.28x
QECTOR OpenCL Batch (GPU, unweighted)	5	10,000	600	0.06000	[0.0555, 0.0648]	179,482	0.56x
QECTOR OpenCL Batch (GPU, weighted)	5	10,000	316	0.03160	[0.0283, 0.0352]	66,580	0.21x
QECTOR Sparse Blossom (CPU)	5	10,000	166	0.01660	[0.0143, 0.0193]	112,331	0.35x
QECTOR Union-Find (CPU)	5	10,000	268	0.02680	[0.0238, 0.0302]	590,577	1.85x
PyMatching v2 (C++)	5	50,000	809	0.01618	[0.0151, 0.0173]	339,590	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	50,000	3041	0.06082	[0.0588, 0.0629]	152,166	0.45x
QECTOR CUDA Batch (GPU, weighted)	5	50,000	1561	0.03122	[0.0297, 0.0328]	67,282	0.20x
QECTOR OpenCL Batch (GPU, unweighted)	5	50,000	3041	0.06082	[0.0588, 0.0629]	140,526	0.41x
QECTOR OpenCL Batch (GPU, weighted)	5	50,000	1561	0.03122	[0.0297, 0.0328]	60,052	0.18x
QECTOR Sparse Blossom (CPU)	5	50,000	810	0.01620	[0.0151, 0.0173]	105,864	0.31x
QECTOR Union-Find (CPU)	5	50,000	1356	0.02712	[0.0257, 0.0286]	672,617	1.98x
PyMatching v2 (C++)	5	100,000	1596	0.01596	[0.0152, 0.0168]	340,202	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	100,000	6094	0.06094	[0.0595, 0.0624]	150,505	0.44x
QECTOR CUDA Batch (GPU, weighted)	5	100,000	3182	0.03182	[0.0307, 0.0329]	69,156	0.20x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR OpenCL Batch (GPU, unweighted)	5	100,000	6094	0.06094	[0.0595, 0.0624]	144,886	0.43x
QECTOR OpenCL Batch (GPU, weighted)	5	100,000	3182	0.03182	[0.0307, 0.0329]	61,421	0.18x
QECTOR Sparse Blossom (CPU)	5	100,000	1596	0.01596	[0.0152, 0.0168]	111,843	0.33x
QECTOR Union-Find (CPU)	5	100,000	2645	0.02645	[0.0255, 0.0275]	694,192	2.04x
PyMatching v2 (C++)	7	1,000	11	0.01100	[0.0062, 0.0196]	87,829	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	1,000	44	0.04400	[0.0329, 0.0586]	24,776	0.28x
QECTOR CUDA Batch (GPU, weighted)	7	1,000	25	0.02500	[0.0170, 0.0366]	4,379	0.05x
QECTOR OpenCL Batch (GPU, unweighted)	7	1,000	44	0.04400	[0.0329, 0.0586]	41,416	0.47x
QECTOR OpenCL Batch (GPU, weighted)	7	1,000	25	0.02500	[0.0170, 0.0366]	5,076	0.06x
QECTOR Sparse Blossom (CPU)	7	1,000	11	0.01100	[0.0062, 0.0196]	17,804	0.20x
QECTOR Union-Find (CPU)	7	1,000	18	0.01800	[0.0114, 0.0283]	85,774	0.98x
PyMatching v2 (C++)	7	5,000	74	0.01480	[0.0118, 0.0185]	95,177	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	5,000	218	0.04360	[0.0383, 0.0496]	68,090	0.71x
QECTOR CUDA Batch (GPU, weighted)	7	5,000	105	0.02100	[0.0174, 0.0254]	16,141	0.17x
QECTOR OpenCL Batch (GPU, unweighted)	7	5,000	218	0.04360	[0.0383, 0.0496]	60,767	0.64x
QECTOR OpenCL Batch (GPU, weighted)	7	5,000	105	0.02100	[0.0174, 0.0254]	10,049	0.11x
QECTOR Sparse Blossom (CPU)	7	5,000	75	0.01500	[0.0120, 0.0188]	20,658	0.22x
QECTOR Union-Find (CPU)	7	5,000	107	0.02140	[0.0177, 0.0258]	147,940	1.55x
PyMatching v2 (C++)	7	10,000	130	0.01300	[0.0110, 0.0154]	97,231	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	10,000	409	0.04090	[0.0372, 0.0450]	64,826	0.67x
QECTOR CUDA Batch (GPU, weighted)	7	10,000	180	0.01800	[0.0156, 0.0208]	13,552	0.14x
QECTOR OpenCL Batch (GPU, unweighted)	7	10,000	409	0.04090	[0.0372, 0.0450]	52,408	0.54x
QECTOR OpenCL Batch (GPU, weighted)	7	10,000	180	0.01800	[0.0156, 0.0208]	8,693	0.09x
QECTOR Sparse Blossom (CPU)	7	10,000	133	0.01330	[0.0112, 0.0157]	20,409	0.21x
QECTOR Union-Find (CPU)	7	10,000	209	0.02090	[0.0183, 0.0239]	153,667	1.58x
PyMatching v2 (C++)	7	50,000	619	0.01238	[0.0114, 0.0134]	96,789	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	50,000	2197	0.04394	[0.0422, 0.0458]	44,880	0.46x
QECTOR OpenCL Batch (GPU, unweighted)	7	50,000	2197	0.04394	[0.0422, 0.0458]	236,797	2.45x
QECTOR Sparse Blossom (CPU)	7	50,000	624	0.01248	[0.0115, 0.0135]	20,334	0.21x
QECTOR Union-Find (CPU)	7	50,000	1055	0.02110	[0.0199, 0.0224]	164,348	1.70x
PyMatching v2 (C++)	7	100,000	1220	0.01220	[0.0115, 0.0129]	97,941	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	100,000	4274	0.04274	[0.0415, 0.0440]	45,353	0.46x
QECTOR OpenCL Batch (GPU, unweighted)	7	100,000	4274	0.04274	[0.0415, 0.0440]	244,318	2.50x
QECTOR Sparse Blossom (CPU)	7	100,000	1233	0.01233	[0.0117, 0.0130]	20,975	0.21x
QECTOR Union-Find (CPU)	7	100,000	2042	0.02042	[0.0196, 0.0213]	173,669	1.77x
PyMatching v2 (C++)	9	1,000	7	0.00700	[0.0034, 0.0144]	38,465	1.00x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR CUDA Batch (GPU, unweighted)	9	1,000	56	0.05600	[0.0434, 0.0720]	10,067	0.26x
QECTOR CUDA Batch (GPU, weighted)	9	1,000	17	0.01700	[0.0106, 0.0271]	1,120	0.03x
QECTOR OpenCL Batch (GPU, unweighted)	9	1,000	56	0.05600	[0.0434, 0.0720]	17,223	0.45x
QECTOR OpenCL Batch (GPU, weighted)	9	1,000	17	0.01700	[0.0106, 0.0271]	1,205	0.03x
QECTOR Sparse Blossom (CPU)	9	1,000	7	0.00700	[0.0034, 0.0144]	2,696	0.07x
QECTOR Union-Find (CPU)	9	1,000	14	0.01400	[0.0084, 0.0234]	17,940	0.47x
PyMatching v2 (C++)	9	5,000	36	0.00720	[0.0052, 0.0100]	42,068	1.00x
QECTOR CUDA Batch (GPU, unweighted)	9	5,000	224	0.04480	[0.0394, 0.0509]	29,601	0.70x
QECTOR CUDA Batch (GPU, weighted)	9	5,000	57	0.01140	[0.0088, 0.0147]	4,495	0.11x
QECTOR OpenCL Batch (GPU, unweighted)	9	5,000	224	0.04480	[0.0394, 0.0509]	25,801	0.61x
QECTOR OpenCL Batch (GPU, weighted)	9	5,000	57	0.01140	[0.0088, 0.0147]	2,371	0.06x
QECTOR Sparse Blossom (CPU)	9	5,000	36	0.00720	[0.0052, 0.0100]	2,719	0.07x
QECTOR Union-Find (CPU)	9	5,000	85	0.01700	[0.0138, 0.0210]	46,387	1.10x
PyMatching v2 (C++)	9	10,000	78	0.00780	[0.0063, 0.0097]	42,397	1.00x
QECTOR CUDA Batch (GPU, unweighted)	9	10,000	431	0.04310	[0.0393, 0.0473]	27,672	0.65x
QECTOR CUDA Batch (GPU, weighted)	9	10,000	140	0.01400	[0.0119, 0.0165]	3,382	0.08x
QECTOR OpenCL Batch (GPU, unweighted)	9	10,000	431	0.04310	[0.0393, 0.0473]	21,899	0.52x
QECTOR OpenCL Batch (GPU, weighted)	9	10,000	140	0.01400	[0.0119, 0.0165]	2,043	0.05x
QECTOR Sparse Blossom (CPU)	9	10,000	79	0.00790	[0.0063, 0.0098]	2,606	0.06x
QECTOR Union-Find (CPU)	9	10,000	156	0.01560	[0.0134, 0.0182]	41,347	0.97x
PyMatching v2 (C++)	9	50,000	450	0.00900	[0.0082, 0.0099]	42,085	1.00x
QECTOR CUDA Batch (GPU, unweighted)	9	50,000	2324	0.04648	[0.0447, 0.0484]	18,273	0.43x
QECTOR OpenCL Batch (GPU, unweighted)	9	50,000	2324	0.04648	[0.0447, 0.0484]	106,560	2.53x
QECTOR Sparse Blossom (CPU)	9	50,000	453	0.00906	[0.0083, 0.0099]	2,649	0.06x
QECTOR Union-Find (CPU)	9	50,000	827	0.01654	[0.0155, 0.0177]	43,803	1.04x
PyMatching v2 (C++)	9	100,000	878	0.00878	[0.0082, 0.0094]	42,295	1.00x
QECTOR OpenCL Batch (GPU, unweighted)	9	100,000	4663	0.04663	[0.0453, 0.0480]	107,529	2.54x
QECTOR Union-Find (CPU)	9	100,000	1732	0.01732	[0.0165, 0.0181]	44,420	1.05x
PyMatching v2 (C++)	11	1,000	7	0.00700	[0.0034, 0.0144]	19,696	1.00x
QECTOR CUDA Batch (GPU, unweighted)	11	1,000	44	0.04400	[0.0329, 0.0586]	5,680	0.29x
QECTOR CUDA Batch (GPU, weighted)	11	1,000	10	0.01000	[0.0054, 0.0183]	330	0.02x
QECTOR OpenCL Batch (GPU, unweighted)	11	1,000	44	0.04400	[0.0329, 0.0586]	7,614	0.39x
QECTOR OpenCL Batch (GPU, weighted)	11	1,000	10	0.01000	[0.0054, 0.0183]	338	0.02x
QECTOR Sparse Blossom (CPU)	11	1,000	7	0.00700	[0.0034, 0.0144]	734	0.04x
QECTOR Union-Find (CPU)	11	1,000	13	0.01300	[0.0076, 0.0221]	3,709	0.19x
PyMatching v2 (C++)	11	5,000	24	0.00480	[0.0032, 0.0071]	21,163	1.00x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR CUDA Batch (GPU, unweighted)	11	5,000	229	0.04580	[0.0403, 0.0520]	16,031	0.76x
QECTOR OpenCL Batch (GPU, unweighted)	11	5,000	229	0.04580	[0.0403, 0.0520]	12,561	0.59x
QECTOR Sparse Blossom (CPU)	11	5,000	24	0.00480	[0.0032, 0.0071]	857	0.04x
QECTOR Union-Find (CPU)	11	5,000	71	0.01420	[0.0113, 0.0179]	8,765	0.41x
PyMatching v2 (C++)	11	10,000	77	0.00770	[0.0062, 0.0096]	21,335	1.00x
QECTOR CUDA Batch (GPU, unweighted)	11	10,000	440	0.04400	[0.0402, 0.0482]	13,621	0.64x
QECTOR OpenCL Batch (GPU, unweighted)	11	10,000	440	0.04400	[0.0402, 0.0482]	10,653	0.50x
QECTOR Sparse Blossom (CPU)	11	10,000	78	0.00780	[0.0063, 0.0097]	890	0.04x
QECTOR Union-Find (CPU)	11	10,000	172	0.01720	[0.0148, 0.0199]	8,294	0.39x
PyMatching v2 (C++)	11	50,000	334	0.00668	[0.0060, 0.0074]	21,598	1.00x
QECTOR OpenCL Batch (GPU, unweighted)	11	50,000	2111	0.04222	[0.0405, 0.0440]	53,218	2.46x
QECTOR Union-Find (CPU)	11	50,000	839	0.01678	[0.0157, 0.0179]	9,722	0.45x
PyMatching v2 (C++)	11	100,000	647	0.00647	[0.0060, 0.0070]	21,511	1.00x
PyMatching v2 (C++)	13	1,000	3	0.00300	[0.0010, 0.0088]	10,351	1.00x
QECTOR CUDA Batch (GPU, unweighted)	13	1,000	39	0.03900	[0.0287, 0.0529]	3,172	0.31x
QECTOR CUDA Batch (GPU, weighted)	13	1,000	8	0.00800	[0.0041, 0.0157]	151	0.01x
QECTOR OpenCL Batch (GPU, unweighted)	13	1,000	39	0.03900	[0.0287, 0.0529]	4,715	0.46x
QECTOR OpenCL Batch (GPU, weighted)	13	1,000	8	0.00800	[0.0041, 0.0157]	166	0.02x
QECTOR Sparse Blossom (CPU)	13	1,000	3	0.00300	[0.0010, 0.0088]	291	0.03x
QECTOR Union-Find (CPU)	13	1,000	11	0.01100	[0.0062, 0.0196]	1,364	0.13x
PyMatching v2 (C++)	13	5,000	18	0.00360	[0.0023, 0.0057]	11,752	1.00x
QECTOR CUDA Batch (GPU, unweighted)	13	5,000	201	0.04020	[0.0351, 0.0460]	9,104	0.78x
QECTOR OpenCL Batch (GPU, unweighted)	13	5,000	201	0.04020	[0.0351, 0.0460]	7,151	0.61x
QECTOR Sparse Blossom (CPU)	13	5,000	19	0.00380	[0.0024, 0.0059]	293	0.03x
QECTOR Union-Find (CPU)	13	5,000	98	0.01960	[0.0161, 0.0238]	1,786	0.15x
PyMatching v2 (C++)	13	10,000	37	0.00370	[0.0027, 0.0051]	11,976	1.00x
QECTOR CUDA Batch (GPU, unweighted)	13	10,000	410	0.04100	[0.0373, 0.0451]	7,634	0.64x
QECTOR OpenCL Batch (GPU, unweighted)	13	10,000	410	0.04100	[0.0373, 0.0451]	25,937	2.17x
QECTOR Union-Find (CPU)	13	10,000	140	0.01400	[0.0119, 0.0165]	2,571	0.21x
PyMatching v2 (C++)	13	50,000	221	0.00442	[0.0039, 0.0050]	12,079	1.00x
PyMatching v2 (C++)	13	100,000	445	0.00445	[0.0041, 0.0049]	12,096	1.00x
PyMatching v2 (C++)	15	1,000	2	0.00200	[0.0005, 0.0073]	5,755	1.00x
QECTOR CUDA Batch (GPU, unweighted)	15	1,000	41	0.04100	[0.0304, 0.0551]	1,935	0.34x
QECTOR OpenCL Batch (GPU, unweighted)	15	1,000	41	0.04100	[0.0304, 0.0551]	2,488	0.43x
QECTOR Sparse Blossom (CPU)	15	1,000	2	0.00200	[0.0005, 0.0073]	73	0.01x
QECTOR Union-Find (CPU)	15	1,000	12	0.01200	[0.0069, 0.0209]	370	0.06x
PyMatching v2 (C++)	15	5,000	12	0.00240	[0.0014, 0.0042]	5,161	1.00x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR CUDA Batch (GPU, unweighted)	15	5,000	170	0.03400	[0.0293, 0.0394]	5,276	1.02x
QECTOR OpenCL Batch (GPU, unweighted)	15	5,000	170	0.03400	[0.0293, 0.0394]	14,111	2.73x
QECTOR Union-Find (CPU)	15	5,000	72	0.01440	[0.0115, 0.0181]	511	0.10x
PyMatching v2 (C++)	15	10,000	44	0.00440	[0.0033, 0.0059]	5,465	1.00x
QECTOR CUDA Batch (GPU, unweighted)	15	10,000	376	0.03760	[0.0340, 0.0415]	4,398	0.81x
QECTOR OpenCL Batch (GPU, unweighted)	15	10,000	376	0.03760	[0.0340, 0.0415]	14,809	2.71x
PyMatching v2 (C++)	15	50,000	171	0.00342	[0.0029, 0.0040]	4,847	1.00x

Not measured (93 cells)

Listed so the gaps are explicit. These cells were never run and carry no numbers; nothing here was extrapolated or synthesised.

Decoder	d	Shots	Reason	Projected
ldpc BP-OSD	3	100,000	over per-cell decode budget	40s
ldpc BP-OSD	5	5,000	over per-cell decode budget	46s
ldpc BP-OSD	5	10,000	over per-cell decode budget	91s
ldpc BP-OSD	5	50,000	over per-cell decode budget	456s
ldpc BP-OSD	5	100,000	over per-cell decode budget	913s
QECTOR CUDA Batch (GPU, weighted)	7	50,000	over per-cell decode budget	39s
QECTOR CUDA Batch (GPU, weighted)	7	100,000	over per-cell decode budget	78s
QECTOR OpenCL Batch (GPU, weighted)	7	50,000	over per-cell decode budget	46s
QECTOR OpenCL Batch (GPU, weighted)	7	100,000	over per-cell decode budget	93s
ldpc BP-OSD	7	1,000	over per-cell decode budget	35s
ldpc BP-OSD	7	5,000	over per-cell decode budget	177s
ldpc BP-OSD	7	10,000	over per-cell decode budget	353s
ldpc BP-OSD	7	50,000	over per-cell decode budget	1,767s
ldpc BP-OSD	7	100,000	over per-cell decode budget	3,534s
QECTOR Sparse Blossom (CPU)	9	100,000	over per-cell decode budget	41s
QECTOR CUDA Batch (GPU, unweighted)	9	100,000	over per-cell decode budget	30s
QECTOR CUDA Batch (GPU, weighted)	9	50,000	over per-cell decode budget	121s
QECTOR CUDA Batch (GPU, weighted)	9	100,000	over per-cell decode budget	241s
QECTOR OpenCL Batch (GPU, weighted)	9	50,000	over per-cell decode budget	119s
QECTOR OpenCL Batch (GPU, weighted)	9	100,000	over per-cell decode budget	238s
ldpc BP-OSD	9	1,000	over per-cell decode budget	98s
ldpc BP-OSD	9	5,000	over per-cell decode budget	492s
ldpc BP-OSD	9	10,000	over per-cell decode budget	985s
ldpc BP-OSD	9	50,000	over per-cell decode budget	4,923s
ldpc BP-OSD	9	100,000	over per-cell decode budget	9,846s
QECTOR Sparse Blossom (CPU)	11	50,000	over per-cell decode budget	72s
QECTOR Sparse Blossom (CPU)	11	100,000	over per-cell decode budget	144s
QECTOR Union-Find (CPU)	11	100,000	over per-cell decode budget	42s
QECTOR CUDA Batch (GPU, unweighted)	11	50,000	over per-cell decode budget	34s
QECTOR CUDA Batch (GPU, unweighted)	11	100,000	over per-cell decode budget	68s
QECTOR CUDA Batch (GPU, weighted)	11	5,000	over per-cell decode budget	40s
QECTOR CUDA Batch (GPU, weighted)	11	10,000	over per-cell decode budget	80s
QECTOR CUDA Batch (GPU, weighted)	11	50,000	over per-cell decode budget	398s
QECTOR CUDA Batch (GPU, weighted)	11	100,000	over per-cell decode budget	795s
QECTOR OpenCL Batch (GPU, unweighted)	11	100,000	over per-cell decode budget	38s
QECTOR OpenCL Batch (GPU, weighted)	11	5,000	over per-cell decode budget	39s
QECTOR OpenCL Batch (GPU, weighted)	11	10,000	over per-cell decode budget	78s
QECTOR OpenCL Batch (GPU, weighted)	11	50,000	over per-cell decode budget	388s
QECTOR OpenCL Batch (GPU, weighted)	11	100,000	over per-cell decode budget	777s
ldpc BP-OSD	11	1,000	over per-cell decode budget	197s
ldpc BP-OSD	11	5,000	over per-cell decode budget	983s
ldpc BP-OSD	11	10,000	over per-cell decode budget	1,965s

Decoder	d	Shots	Reason	Projected
ldpc BP-OSD	11	50,000	over per-cell decode budget	9,827s
ldpc BP-OSD	11	100,000	over per-cell decode budget	19,654s
QECTOR Sparse Blossom (CPU)	13	10,000	over per-cell decode budget	39s
QECTOR Sparse Blossom (CPU)	13	50,000	over per-cell decode budget	196s
QECTOR Sparse Blossom (CPU)	13	100,000	over per-cell decode budget	392s
QECTOR Union-Find (CPU)	13	50,000	over per-cell decode budget	66s
QECTOR Union-Find (CPU)	13	100,000	over per-cell decode budget	131s
QECTOR CUDA Batch (GPU, unweighted)	13	50,000	over per-cell decode budget	59s
QECTOR CUDA Batch (GPU, unweighted)	13	100,000	over per-cell decode budget	119s
QECTOR CUDA Batch (GPU, weighted)	13	5,000	over per-cell decode budget	82s
QECTOR CUDA Batch (GPU, weighted)	13	10,000	over per-cell decode budget	163s
QECTOR CUDA Batch (GPU, weighted)	13	50,000	over per-cell decode budget	816s
QECTOR CUDA Batch (GPU, weighted)	13	100,000	over per-cell decode budget	1,632s
QECTOR OpenCL Batch (GPU, unweighted)	13	50,000	over per-cell decode budget	33s
QECTOR OpenCL Batch (GPU, unweighted)	13	100,000	over per-cell decode budget	67s
QECTOR OpenCL Batch (GPU, weighted)	13	5,000	over per-cell decode budget	79s
QECTOR OpenCL Batch (GPU, weighted)	13	10,000	over per-cell decode budget	159s
QECTOR OpenCL Batch (GPU, weighted)	13	50,000	over per-cell decode budget	795s
QECTOR OpenCL Batch (GPU, weighted)	13	100,000	over per-cell decode budget	1,589s
ldpc BP-OSD	13	1,000	over per-cell decode budget	351s
ldpc BP-OSD	13	5,000	over per-cell decode budget	1,753s
ldpc BP-OSD	13	10,000	over per-cell decode budget	3,505s
ldpc BP-OSD	13	50,000	over per-cell decode budget	17,526s
ldpc BP-OSD	13	100,000	over per-cell decode budget	35,053s
QECTOR Sparse Blossom (CPU)	15	5,000	over per-cell decode budget	74s
QECTOR Sparse Blossom (CPU)	15	10,000	over per-cell decode budget	147s
QECTOR Sparse Blossom (CPU)	15	50,000	over per-cell decode budget	737s
QECTOR Sparse Blossom (CPU)	15	100,000	over per-cell decode budget	1,474s
QECTOR Union-Find (CPU)	15	10,000	over per-cell decode budget	38s
QECTOR Union-Find (CPU)	15	50,000	over per-cell decode budget	190s
QECTOR Union-Find (CPU)	15	100,000	over per-cell decode budget	380s
QECTOR CUDA Batch (GPU, unweighted)	15	50,000	over per-cell decode budget	101s
QECTOR CUDA Batch (GPU, unweighted)	15	100,000	over per-cell decode budget	202s
QECTOR CUDA Batch (GPU, weighted)	15	1,000	over per-cell decode budget	36s
QECTOR CUDA Batch (GPU, weighted)	15	5,000	over per-cell decode budget	182s
QECTOR CUDA Batch (GPU, weighted)	15	10,000	over per-cell decode budget	364s
QECTOR CUDA Batch (GPU, weighted)	15	50,000	over per-cell decode budget	1,821s
QECTOR CUDA Batch (GPU, weighted)	15	100,000	over per-cell decode budget	3,641s
QECTOR OpenCL Batch (GPU, unweighted)	15	50,000	over per-cell decode budget	60s
QECTOR OpenCL Batch (GPU, unweighted)	15	100,000	over per-cell decode budget	120s
QECTOR OpenCL Batch (GPU, weighted)	15	1,000	over per-cell decode budget	36s
QECTOR OpenCL Batch (GPU, weighted)	15	5,000	over per-cell decode budget	179s
QECTOR OpenCL Batch (GPU, weighted)	15	10,000	over per-cell decode budget	357s

Decoder	d	Shots	Reason	Projected
QECTOR OpenCL Batch (GPU, weighted)	15	50,000	over per-cell decode budget	1,785s
QECTOR OpenCL Batch (GPU, weighted)	15	100,000	over per-cell decode budget	3,571s
PyMatching v2 (C++)	15	100,000	over per-cell decode budget	33s
ldpc BP-OSD	15	1,000	over per-cell decode budget	564s
ldpc BP-OSD	15	5,000	over per-cell decode budget	2,821s
ldpc BP-OSD	15	10,000	over per-cell decode budget	5,641s
ldpc BP-OSD	15	50,000	over per-cell decode budget	28,205s
ldpc BP-OSD	15	100,000	over per-cell decode budget	56,411s